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WOMEN SAFETY APP – SOS AND LIVE TRACKING

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Abstract

This system is a mobile application that enhances personal safety for women through an instant SOS alert and live location tracking mechanism. By triggering a single SOS button, the app immediately sends the user's real-time GPS coordinates and an emergency alert to pre-selected trusted contacts. Traditional safety measures such as manual calls can be slow and unreliable during a crisis. This project aims to reduce response time and remove the need for the user to explain the situation while under threat. The live tracking feature continuously shares the user's movement with guardians until the emergency is resolved, improving transparency. This helps families and responders locate the user quickly, promoting faster intervention. The system also improves accessibility by working with minimal interaction during high-stress situations. Ultimately, the app enhances women's safety, leading to quicker response and greater peace of mind.

Introduction

The Women Safety App is a mobile-based solution built around two core features: an instant SOS alert and continuous live location tracking. When a user feels unsafe, a single tap on the SOS button sends her real-time GPS location and an emergency message to pre-selected trusted contacts, removing the delay and difficulty of making a phone call while under threat. Once triggered, the app keeps sharing the user's live location so that guardians can follow her movement until she is safe. This approach simplifies emergency response, making help easier to reach and faster to arrive. It reduces dependence on bystanders or manual communication during a crisis, giving users a more reliable and accessible safety option. Overall, the app aims to improve response time, increase a sense of security, and make everyday travel safer for women.

Existing System

Most current personal-safety approaches rely on manually calling or messaging a family member or friend when trouble arises, which takes precious time and requires a clear head during a stressful situation. Some existing safety apps offer SOS alerts but lack continuous live tracking, sending only a one-time location instead of ongoing movement updates. Others depend on a stable internet connection or a complex sequence of taps, making them difficult to use in a panic. Many solutions also fail to integrate emergency contacts and location sharing into a single, seamless action, forcing the user to switch between multiple apps. These gaps often lead to delayed response, inaccurate location information, and reduced chances of timely intervention, highlighting the need for a faster and more dependable system.

Proposed System

The proposed Women Safety App addresses these gaps with a single-tap SOS button that instantly sends the user's real-time GPS coordinates and an emergency alert to trusted contacts. Unlike existing solutions, it also enables continuous live location tracking, allowing guardians to monitor the user's movement until the emergency is resolved rather than seeing only a one-time location. The app is designed for minimal interaction during high-stress moments, reducing the number of steps needed to raise an alert. It works reliably even with limited connectivity by prioritizing the emergency alert transmission. By combining instant alerting with ongoing tracking in one streamlined app, the proposed system aims to cut response time, improve location accuracy, and give both users and their families greater confidence and peace of mind.

System Architecture

The system follows a client-server architecture with three main layers. The mobile application layer captures the SOS trigger and continuously reads the device's GPS location using location services. The communication layer transmits this data over the internet to a backend server, which authenticates the user, processes the alert, and stores live location updates. The backend then pushes real-time notifications and location links to the trusted contacts' devices through a messaging or push-notification service. A database maintains user profiles, emergency contact lists, and location history for the duration of an active alert. This layered design keeps the SOS trigger fast and lightweight on the user's device while the server handles contact notification, data storage, and live tracking updates in the background.

System Models

The system is modeled using a Use Case diagram showing the primary user actions: register, manage emergency contacts, trigger SOS, and share live location. A Data Flow Diagram illustrates how location data moves from the mobile app through the backend server to the trusted contacts' devices. A Sequence diagram captures the timeline of an SOS event, from the button trigger through location capture, server processing, and contact notification. Together, these models describe the interactions between the user, the mobile application, the backend server, and the emergency contacts, giving a clear picture of how the system behaves during both normal use and an active emergency.

Requirements Specifications

Functional requirements include: user registration and login, adding and managing trusted emergency contacts, a one-tap SOS trigger, real-time GPS location capture, continuous live location sharing during an active alert, and automated alert notifications to trusted contacts. Non-functional requirements include fast response time for the SOS trigger, reliable performance under low-connectivity conditions, data privacy and secure storage of location and contact information, and an intuitive interface usable under stress. Hardware requirements include a smartphone with GPS and internet connectivity. Software requirements include an Android/iOS development environment, a backend server with a database, and mapping and location APIs for real-time tracking and notification delivery.

Conclusion

The Women Safety App provides a practical and accessible way to improve personal safety through an instant SOS alert combined with continuous live location tracking. By reducing the number of steps needed to raise an alarm and keeping trusted contacts updated in real time, the system aims to shorten emergency response time and improve the chances of timely help. The proposed architecture is lightweight on the user's device while still supporting reliable alerting and tracking through the backend. Future enhancements could include direct integration with local police control rooms, voice-activated SOS triggers, and offline alert queuing for areas with poor connectivity, further strengthening the app's reliability and reach.